

DATABASE MANAGEMENT SYSTEMS (IT 2040)

Question 1

A relation stores employee and department data together. When deleting the last employee from a department results in losing all department information, this is an example of:

- A) Insertion anomaly
- B) Deletion anomaly
- C) Update anomaly
- D) Referential integrity violation

Answer: B

Question 2

Consider relation $R(A, B, C, D)$ with FDs: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$. What is A^+ (attribute closure of A)?

- A) $\{A\}$
- B) $\{A, B\}$
- C) $\{A, B, C\}$
- D) $\{A, B, C, D\}$

Answer: D

Question 3

Which Armstrong's Axiom states: "If $X \supseteq Y$, then $Y \rightarrow X$ "?

- A) Reflexivity
- B) Augmentation
- C) Transitivity
- D) Union

Answer: A

Question 4

A decomposition has the lossless join property if:

- A) No data is lost during decomposition
- B) Joining the decomposed relations produces exactly the original relation

- C) All functional dependencies are preserved
- D) No NULL values are introduced

Answer: B

Question 5

Given $R(A, B, C, D, E)$ with FDs: $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E\}$. What is the primary key?

- A) A
- B) AB
- C) ABC
- D) ABCDE

Answer: B

Question 6

A relation is in 1NF if:

- A) It has no partial dependencies
- B) It has no transitive dependencies
- C) All attributes contain only atomic (indivisible) values
- D) Every determinant is a candidate key

Answer: C

Question 7

Consider $R(A, B, C, D)$ with key AB and FD: $A \rightarrow C$. This relation violates:

- A) 1NF only
- B) 2NF (has partial dependency)
- C) 3NF (has transitive dependency)
- D) BCNF only

Answer: B

Question 8

A functional dependency $X \rightarrow Y$ is a FULL functional dependency if:

- A) X is a superkey
- B) Y contains all attributes of the relation
- C) Removing any attribute from X means the dependency no longer holds
- D) X and Y have no common attributes

Answer: C

Question 9

Given $R(A, B, C, D)$ with FDs: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$. How many candidate keys does R have?

- A) 1
- B) 2
- C) 4
- D) Cannot be determined

Answer: C

Question 10

A relation R is in 3NF if:

- A) It is in 2NF and has no transitive dependencies among nonprime attributes
- B) It has no partial dependencies
- C) Every determinant is a candidate key
- D) It is in 1NF only

Answer: A

Question 11

Consider $R(A, B, C)$ with FDs: $\{A \rightarrow B, B \rightarrow C\}$. To normalize to 3NF, decompose into:

- A) $R_1(A, B), R_2(B, C)$
- B) $R_1(A, C), R_2(B, C)$
- C) $R_1(A, B, C)$ - already in 3NF
- D) $R_1(A), R_2(B), R_3(C)$

Answer: A

Question 12

A relation is in BCNF if for every nontrivial FD $X \rightarrow A$:

- A) X is a prime attribute
- B) A is a prime attribute
- C) X is a superkey
- D) $X \rightarrow A$ is a full dependency

Answer: C

Question 13

The dependency preserving property ensures that:

- A) No data is lost during JOIN operations
- B) Constraints on the original relation can be enforced by constraints on decomposed relations
- C) All candidate keys are preserved
- D) The decomposition is lossless

Answer: B

Question 14

Given $R(A, B, C, D)$ with FDs: $\{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. Which FD violates BCNF?

- A) $AB \rightarrow C$ (AB is a key)
- B) $C \rightarrow D$ (C is not a superkey)
- C) $D \rightarrow A$ (D is not a superkey)
- D) Both B and C

Answer: D

Question 15

In relation $R(ABCDEF)$ with primary key AB and FD: $C \rightarrow D$, what type of anomaly occurs?

- A) No anomaly - relation is properly designed
- B) Insertion anomaly when adding new C-D pair without knowing AB
- C) Update anomaly when C value appears multiple times
- D) Both B and C

Answer: D

Question 16

A prime attribute is:

- A) The primary key of a relation
- B) An attribute that appears in the left side of any FD
- C) An attribute that is a member of any candidate key
- D) The first attribute in the relation schema

Answer: C

Question 17

Consider $R(A, B, C, D)$ with FDs: $\{A \rightarrow BC, C \rightarrow D\}$. What is the highest normal form?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

Answer: B

Question 18

Using augmentation axiom, if $X \rightarrow Y$, then for any Z:

- A) $XZ \rightarrow Y$
- B) $X \rightarrow YZ$
- C) $XZ \rightarrow YZ$
- D) $Z \rightarrow Y$

Answer: C

Question 19

Which decomposition property is MORE important for database integrity?

- A) Lossless join property
- B) Dependency preserving property
- C) Both are equally important
- D) Neither affects integrity

Answer: A

Question 20

Given $R(A, B, C, D, E)$ with FDs: $\{A \rightarrow B, BC \rightarrow E, ED \rightarrow A\}$. To find if BC is a key, we compute:

- A) B^+ and C^+

- B) (BC)+
- C) A+
- D) E+

Answer: B

SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply)

Question 21

Redundancy in a poorly designed relation can cause: **(Select all that apply)**

- A) Insertion anomalies
- B) Deletion anomalies
- C) Update anomalies
- D) Faster query performance
- E) Wasted storage space

Answers: A, B, C, E

Question 22

Which are valid Armstrong's Axioms? **(Select all that apply)**

- A) Reflexivity
- B) Augmentation
- C) Transitivity
- D) Union
- E) Decomposition

Answers: A, B, C

Question 23

During normalization from 1NF to BCNF: **(Select all that apply)**

- A) Relations always decrease in number

- B) Redundancy decreases
- C) Query complexity may increase
- D) Some functional dependencies might be lost
- E) The number of NULL values may increase or decrease

Answers: B, C, D, E

Question 24

Given $R(A, B, C)$ with FDs: $\{A \rightarrow B, B \rightarrow C\}$, which are true? **(Select all that apply)**

- A) A is a candidate key
- B) There is a transitive dependency
- C) The relation is in 2NF
- D) The relation is in 3NF
- E) The relation is in BCNF

Answers: A, B, C

Question 25

Which statements about BCNF are correct? **(Select all that apply)**

- A) Every BCNF relation is in 3NF
- B) Every 3NF relation is in BCNF
- C) BCNF is stricter than 3NF
- D) BCNF allows some transitive dependencies
- E) BCNF decomposition may not preserve dependencies

Answers: A, C, E

Question 26

A relation in 3NF but not BCNF must have: **(Select all that apply)**

- A) At least two candidate keys
- B) A determinant that is not a superkey but determines a prime attribute

- C) Overlapping candidate keys
- D) Transitive dependencies involving only prime attributes
- E) Composite candidate keys

Answers: B, C

Question 27

Which can be determined using attribute closure? **(Select all that apply)**

- A) Whether a set of attributes is a superkey
- B) Whether a set of attributes is a candidate key
- C) All FDs that hold in the relation
- D) Whether an FD is implied by a set of FDs
- E) The normal form of a relation

Answers: A, B, D

Question 28

To normalize a relation from 2NF to 3NF, remove: **(Select all that apply)**

- A) Partial dependencies
- B) Transitive dependencies where nonprime determines nonprime
- C) All FDs where determinant is not a key
- D) Multivalued attributes
- E) Composite attributes

Answers: B

Question 29

A relation $R(A, B, C, D)$ is in BCNF if: **(Select all that apply)**

- A) It is in 3NF
- B) For every FD $X \rightarrow A$, either X is a superkey or A is prime
- C) For every FD $X \rightarrow A$, X is a superkey (nontrivial FDs)
- D) It has no redundancy

- E) It has no partial or transitive dependencies

Answers: A, C, E

Question 30

Insertion anomalies can occur when: **(Select all that apply)**

- A) Cannot insert data without knowing unrelated information
- B) Required fields are NULL
- C) Redundant data must be entered consistently
- D) Primary key cannot be determined
- E) Foreign key constraints are violated

Answers: A, C

SECTION C: SCENARIO-BASED QUESTIONS

Question 31

Scenario: Consider relation COURSE_ENROLLMENT(StudentID, StudentName, CourseID, CourseName, InstructorID, InstructorName, Grade)

- FDs: {StudentID → StudentName, CourseID → CourseName, CourseID → InstructorID, InstructorID → InstructorName, StudentID, CourseID → Grade}

What is the primary key?

- A) StudentID
- B) CourseID
- C) (StudentID, CourseID)
- D) (StudentID, CourseID, InstructorID)

Answer: C

Question 32

Continuing from Q31, what is the highest normal form of COURSE_ENROLLMENT?

- A) 1NF
- B) 2NF

- C) 3NF
- D) BCNF

Answer: A

Question 33

Continuing from Q31, decompose to 3NF. Which decomposition is correct?

- A) R1(StudentID, StudentName), R2(CourseID, CourseName, InstructorID), R3(InstructorID, InstructorName), R4(StudentID, CourseID, Grade)
- B) R1(StudentID, StudentName), R2(CourseID, CourseName, InstructorID, InstructorName), R3(StudentID, CourseID, Grade)
- C) R1(StudentID, StudentName, CourseID, CourseName), R2(InstructorID, InstructorName, Grade)
- D) R1(StudentID, StudentName), R2(CourseID, CourseName, InstructorName), R3(StudentID, CourseID, Grade)

Answer: B

Question 34

Scenario: R(A, B, C, D, E) with FDs: {A → B, B → C, C → D, D → E, E → A}

Which statements are true?

- A) Every single attribute is a candidate key
- B) The relation is in BCNF
- C) The relation has redundancy
- D) A decomposition is needed

Answer: A

Question 35

Scenario: RENTAL(PropertyID, CountyName, Lot#, Area, Price, TaxRate) with FDs:

- FD1: PropertyID → CountyName, Lot#, Area, Price, TaxRate
- FD2: CountyName, Lot# → PropertyID, Area, Price, TaxRate
- FD3: CountyName → TaxRate
- FD4: Area → Price

What should be the first decomposition step to move toward BCNF?

- A) Separate based on FD3: R1(CountyName, TaxRate), R2(PropertyID, CountyName, Lot#, Area, Price)

- B) Separate based on FD4: R1(Area, Price), R2(PropertyID, CountyName, Lot#, Area, TaxRate)
- C) Separate based on FD1: R1(PropertyID, CountyName, Lot#, Area, Price, TaxRate)
- D) No decomposition needed

Answer: A

Question 36

Scenario: EMPLOYEE(EmpID, ProjectID, Hours, EmpName, ProjectName, Location) with FDs:

- EmpID, ProjectID $\rightarrow$ Hours
- EmpID $\rightarrow$ EmpName
- ProjectID $\rightarrow$ ProjectName, Location

Primary key is (EmpID, ProjectID). Normalize to BCNF showing all steps.

Which relations result from complete BCNF normalization?

- A) R1(EmpID, EmpName), R2(ProjectID, ProjectName, Location), R3(EmpID, ProjectID, Hours)
- B) R1(EmpID, EmpName, ProjectID, Hours), R2(ProjectID, ProjectName, Location)
- C) R1(EmpID, ProjectID, Hours, EmpName, ProjectName, Location)
- D) R1(EmpID, EmpName), R2(ProjectID, ProjectName), R3(ProjectID, Location), R4(EmpID, ProjectID, Hours)

Answer: A

Question 37

Scenario: R(A, B, C, D) with FDs: {AB $\rightarrow$ C, C $\rightarrow$ D, D $\rightarrow$ B}

What is the minimal key?

- A) AB
- B) AC
- C) AD
- D) A

Answer: C

Question 38

Scenario: SUPPLIER_PART(SupplierID, PartID, ProjectID, Quantity, SupplierName, PartName, City)

- SupplierID $\rightarrow$ SupplierName, City
- PartID $\rightarrow$ PartName
- SupplierID, PartID, ProjectID $\rightarrow$ Quantity

After normalizing to 3NF, how many relations are created?

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C

Question 39

Scenario: You have R(A, B, C, D, E, F) with AB as primary key and these FDs:

- AB $\rightarrow$ CDEF
- C $\rightarrow$ D
- E $\rightarrow$ F

Decompose to BCNF. Which violates BCNF first?

- A) $AB \rightarrow CDEF$
- B) $C \rightarrow D$
- C) $E \rightarrow F$
- D) Both B and C

Answer: D

Question 40

Scenario: After normalizing a relation to BCNF, you discover that a critical constraint "total project hours must not exceed 160" cannot be checked without joining multiple tables.

What property has been lost?

- A) Lossless join property
- B) Dependency preservation property
- C) Domain constraint enforcement
- D) Atomicity

Answer: B